Example 6.29

Find the acute angle between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + t(2\hat{i} + 2\hat{j} + \hat{k})$ and the straight line passing through the points $(5,1,4)$ and $(9,2,12)$.

Solution

We know that the line $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + t(2\hat{i} + 2\hat{j} + \hat{k})$ is parallel to the vector $2\hat{i} + 2\hat{j} + \hat{k}$.

Direction ratios of the straight line joining the two given points $(5,1,4)$ and $(9,2,12)$ are $4,1,8$ and hence this line is parallel to the vector $4\hat{i} + \hat{j} + 8\hat{k}$.

Therefore, the acute angle between the given two straight lines is

$$\theta = \cos^{-1}\left(\frac{|\vec{b}\cdot\vec{d}|}{|\vec{b}||\vec{d}|}\right), \text{ where } \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k} \text{ and } \vec{d} = 4\hat{i} + \hat{j} + 8\hat{k}.$$

$$\text{Therefore, } \theta = \cos^{-1}\left(\frac{|(2\hat{i} + 2\hat{j} + \hat{k})\cdot(4\hat{i} + \hat{j} + 8\hat{k})|}{|2\hat{i} + 2\hat{j} + \hat{k}||4\hat{i} + \hat{j} + 8\hat{k}|}\right) = \cos^{-1}\left(\frac{2}{3}\right).$$

Example 6.30

Find the acute angle between the straight lines $\frac{x-4}{2} = \frac{y}{1} = \frac{z+1}{-2}$ and $\frac{x-1}{4} = \frac{y+1}{-4} = \frac{z-2}{2}$ and state whether they are parallel or perpendicular.

Solution

Comparing the given lines with the general Cartesian equations of straight lines,

$$\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3} \text{ and } \frac{x-x_2}{d_1} = \frac{y-y_2}{d_2} = \frac{z-z_2}{d_3}$$

we find $(b_1, b_2, b_3) = (2,1,-2)$ and $(d_1, d_2, d_3) = (4,-4,2)$. Therefore, the acute angle between the two straight lines is

$$\theta = \cos^{-1}\left(\frac{|(2)(4) + (1)(-4) + (-2)(2)|}{\sqrt{2^2 + 1^2 + (-2)^2}\sqrt{4^2 + (-4)^2 + 2^2}}\right) = \cos^{-1}(0) = \frac{\pi}{2}$$

Thus the two straight lines are perpendicular.

Example 6.31

Show that the straight line passing through the points $A(6,7,5)$ and $B(8,10,6)$ is perpendicular to the straight line passing through the points $C(10,2,-5)$ and $D(8,3,-4)$.

Solution

The straight line passing through the points $A(6,7,5)$ and $B(8,10,6)$ is parallel to the vector $\vec{b} = \overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 2\hat{i} + 3\hat{j} + \hat{k}$ and the straight line passing through the points $C(10,2,-5)$ and $D(8,3,-4)$ is parallel to the vector $\vec{d} = \overrightarrow{CD} = -2\hat{i} + \hat{j} + \hat{k}$. Therefore, the angle between the two straight lines is the angle between the two vectors $\vec{b}$ and $\vec{d}$. Since

$$\vec{b}\cdot\vec{d} = (2\hat{i} + 3\hat{j} + \hat{k})\cdot(-2\hat{i} + \hat{j} + \hat{k}) = 0.$$

the two vectors are perpendicular, and hence the two straight lines are perpendicular.

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